

Machine Learning and Natural Language Processing Fall 2019

Homework 7

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Project Report

Program function

Given a list of filenames in a directory and a user input string, the `bow.py` program accesses each file sequentially and constructs a bag-of-words dataframe for every dialogue option in said file. Next, the program selects the ten (10) closest dialogues -as in, the dialogues with the highest dot product with regards to the user input- from each file and stores them in a list. The list is updated with every iteration, making sure that the final version consists of the ten best results overall. This sequential approach, instead of a simultaneous bag-of-words construction for all files, was chosen in order to prevent memory errors, due to the limited computational resources at my disposal.

Program results

For user input ‘**Could you please help me?**’ the `bow.py` program produces the following results:
here are the 10 closest dialogues:

```
HOW_TO_BASIC.txt, 0148: Hello how may I help you? Can you help me out with the internet? I..
HOW_TO_BASIC.txt, 0262: Hello how may I help you? I want to know how to search on an online..
HOW_TO_BASIC.txt, 0684: Hello how may I help you? Hello, I need help on how to do something..
HOW_TO_BASIC.txt, 0723: Hello how may I help you? Hi I need you to tell me how to order a..
HOW_TO_BASIC.txt, 0765: Hello how may I help you? i need help What can I help you with today..
HOW_TO_BASIC.txt, 0878: Hello how may I help you? I need help with something Sure, I can help..
HOW_TO_BASIC.txt, 0989: Hello how may I help you? I am new to america and i would like to..
HOW_TO_BASIC.txt, 0963: Hello how may I help you? How do i search for something on a search..
HOW_TO_BASIC.txt, 1020: Hello how may I help you? I want to know how to order a pizza over..
PET_ADVICE.txt, 420: Hello how may I help you? I need help with a pet. Sure. What would..
```

```
total dialogues: 37884
here are the 10 closest dialogues:
HOW_TO_BASIC.txt, 0148: Hello how may I help you? Can you help me out with the internet? I can try. What do you need help with? I need help understanding how to use a search engine. Sure, I should be c
HOW_TO_BASIC.txt, 0262: Hello how may I help you? I want to know how to search on an online search engine. Sure, I can help you with how to use a search engine. Okay, what do I do first? Do you have ar
HOW_TO_BASIC.txt, 0684: Hello how may I help you? Hello, I need help on how to do something. Sure. What can I help with? I need help ordering something over the phone. Alright. I can help you. What are
HOW_TO_BASIC.txt, 0723: Hello how may I help you? Hi I need you to tell me how to order a pizza over the phone. Sure. Where would you like to order from? Well I just need to know how to do it like...I'll
HOW_TO_BASIC.txt, 0765: Hello how may I help you? i need help What can I help you with today? How to order pizza via phone? What restaurant would you like to order pizza? any pizza store Is Pizza Hut c
HOW_TO_BASIC.txt, 0878: Hello how may I help you? I need help with something Sure, I can help you. What do you need? I'm trying to figure out how to find stuff on the internet. Sure, I can help you with
HOW_TO_BASIC.txt, 0989: Hello how may I help you? I am new to america and i would like to order a pizza. Sure, I can help with that. Do you know which pizza place you want to order from? Yes BigPizza I
HOW_TO_BASIC.txt, 0963: Hello how may I help you? How do i search for something on a search engine You can type in google.com and at the top of the page enter what you would like to search. i need extr
HOW_TO_BASIC.txt, 1020: Hello how may I help you? I want to know how to order a pizza over the phone. Is this your first time doing that? Yes. I am afraid of listening to other people speak. Okay get c
PET_ADVICE.txt, 420: Hello how may I help you? I need help with a pet. Sure. What would you need help with? What do you do with a dog? What specifically are you trying to ask about a dog? How do you cr
```

Figure 1: Results

The program also prints the bag-of-words data structures for each file on the screen, while they're being procedurally generated. For instance, Figure 2 showcases the bag-of-words output, in dataframe form, for the `AGREEMENT_BOT.txt` file. More specifically, the transposed version of the original dataframe (Fig 3) is particularly useful for calculating the dot product of each dialogue. The result of said operation is later used to assess the similarity between the user input and various dialogues.

```

please enter your question: Could you please help me?
      hello  how  may   i   ...  snow  faked  maple  syrup
AGREEMENT_BOT_000      1   1   1   9   ...    0    0    0    0
AGREEMENT_BOT_001      1   1   1  12   ...    0    0    0    0
AGREEMENT_BOT_002      1   1   1   8   ...    0    0    0    0
AGREEMENT_BOT_003      1   1   1   5   ...    0    0    0    0
AGREEMENT_BOT_004      1   1   1   9   ...    0    0    0    0

[5 rows x 2781 columns]

```

Figure 2: The original `AGREEMENT_BOT.txt` bow dataframe

```

      AGREEMENT_BOT_000  AGREEMENT_BOT_001  ...  AGREEMENT_BOT_506  input
hello                  1                  1   ...                  1    0
how                   1                  1   ...                  1    0
may                   1                  1   ...                  1    0
i                     9                  12   ...                  5    0
help                  1                  1   ...                  1    1

[5 rows x 508 columns]

```

Figure 3: The transposed `AGREEMENT_BOT.txt` dataframe

Further Testing

Additional examples of the `bow.py` program output can be found below:

‘Football score, please!’

```

AUTO_SORT.txt, 242:  Hello how may I help you?  Can you sort the numbers 1, 5, 32, 53, 25..
CITY_INFO.txt, 201:  Hello how may I help you?  Who founded rome?  According to legend, Ancient..
HOW_TO_BASIC.txt, 0852:  Hello how may I help you?  Hi, I need some help with cooking since..
HOW_TO_BASIC.txt, 0148:  Hello how may I help you?  Can you help me out with the internet?  I..
HOW_TO_BASIC.txt, 0878:  Hello how may I help you?  I need help with something Sure, I can help..
HOW_TO_BASIC.txt, 0989:  Hello how may I help you?  I am new to america and i would like to..
HOW_TO_BASIC.txt, 0176:  Hello how may I help you?  Hi, I need a bit of help with instructions..
HOW_TO_BASIC.txt, 0885:  Hello how may I help you?  I would like some advice on how to order..
HOW_TO_BASIC.txt, 0900:  Hello how may I help you?  Hi, I need help with something.  Sure!  What..
AGREEMENT_BOT.txt, 162:  Hello how may I help you?  I am so great YES!!!!!!!  Everything I..

```

Programmer comments: The above results are likely due to the high frequencies of punctuation tokens such as commas (,) and exclamation points (!) in the specific dialogues.

‘Say something’

```
MUSIC_SUGGESTER.txt, 116: Hello how may I help you? So, I'm having this whole wedding thing..
HOW_TO_BASIC.txt, 0064: Hello how may I help you? I'm trying to search for something online..
HOW_TO_BASIC.txt, 0684: Hello how may I help you? Hello, I need help on how to do something..
HOW_TO_BASIC.txt, 1079: Hello how may I help you? Hi I've never talked to a bot before. What..
PROMPT_GENERATOR.txt, 212: Hello how may I help you? I need some help with something creative..
HOW_TO_BASIC.txt, 0136: Hello how may I help you? I need you to tell me how to look up something..
HOW_TO_BASIC.txt, 0883: Hello how may I help you? hello, I'm trying to search for something..
MUSIC_SUGGESTER.txt, 000: Hello how may I help you? help me with music What can I do for..
NAME_SUGGESTER.txt, 068: Hello how may I help you? I have so much on my mind and I forgot..
NAME_SUGGESTER.txt, 339: Hello how may I help you? help me name something Ok, what are you..
```

Programmer comments: It's worth mentioning here that the label numbering for each dialogue starts at zero (0), rather than one (1).

‘Temperature Rain Thunderstorms Forecast’

```
HOME_BOT.txt, 0466: Hello how may I help you? I want to know what the current house temperature..
WEATHER_CHECK.txt, 452: Hi. How can I help you? Hi. I am wondering if it is going to rain..
HOME_BOT.txt, 0512: Hello how may I help you? what is the set house temperature? Your temperature..
WEATHER_CHECK.txt, 010: Hello how may I help you? Can you help me with the weather Sure..
HOME_BOT.txt, 0668: Hello how may I help you? Can you change the temperature? The current..
HOME_BOT.txt, 0779: Hello how may I help you? I need help managing my home I can help you..
WEATHER_CHECK.txt, 206: weatherbot hello! give me the weather forecast for today sunshine..
WEATHER_CHECK.txt, 365: Hello how may I help you? need to know if the thunderstorm is still..
HOME_BOT.txt, 0958: Hello how may I help you? I need to change the temperature. What would..
WEATHER_CHECK.txt, 071: Hello, how may I help you? How hot will it get on Friday? It will..
```

Programmer comments: As expected, the majority of results are from the `WEATHER_CHECK.txt` file.

Possible Improvements

The `bow.py` program can be further improved with the use of various normalisation techniques for the `.txt` file and input data. Some of those techniques include:

a. Stop-word filtering.

Removing words that do not add any substantial meaning to a text, is often a good way to ensure higher relevance and noise reduction when constructing a bag-of-words dictionary. Utilising a stop-words list, like the one provided by the `nltk` library, in order to remove unnecessary information, would be a good way to increase the efficacy and accuracy of the `bow.py` program's output.

b. The tf-idf metric.

By using the tf-idf weighing method for each term instead of its simple frequency, we can ensure that information-rich words factor more heavily into the dot product operation. The `TfidfVectorizer` function found in `sklearn` is a great tool to use for this specific purpose.